COMPLEX NUMBER & FUNCTIONS

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Complex Numbers - Algebra

A complex number, z is defined to be

$$z = x + iy = (x, y)$$

$$x = \mathrm{Re}(z), \qquad y = \mathrm{Im}(z)$$

$(0, 1)$ Is called the imaginary unit denoted by, i

$$i = \sqrt{-1}, \; i^2 = -1$$

- Two complex numbers, $x + iy$ and $u + iv$ are equal if and only if

$$x = u, \qquad y = v$$

  i.e., if real parts are equal, and imaginary parts are equal

- Let us consider two complex numbers, $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$

1. Addition

$$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$$

$$= (x_1 + x_2) + i(y_1 + y_2)$$

2. Subtraction

$$z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2)$$
$$= (x_1 - x_2) + i(y_1 - y_2)$$

3. Multiplication

$$z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2)$$
$$= x_1 x_2 + ix_1 y_2 + ix_2 y_1 + i^2 y_1 y_2$$
$$= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$$

4. Division, $z_2 \neq 0$

$$\frac{z_1}{z_2} = \frac{(x_1 + iy_1)}{(x_2 + iy_2)}$$
$$= \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)}$$
$$= \frac{(x_1 x_2 + y_1 y_2) + i(-x_1 y_2 + x_2 y_1)}{x_2^2 + y_2^2}$$
$$= \frac{(x_1 x_2 + y_1 y_2)}{x_2^2 + y_2^2} + i\frac{(-x_1 y_2 + x_2 y_1)}{x_2^2 + y_2^2}$$

Example

Given, $z_1 = 8 + 3i,\ \ z_2 = 9 - 2i$. Calculate, $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, $\frac{z_1}{z_2}$

$$z_1 + z_2 = (8 + 3i) + (9 - 2i)$$
$$= (8 + 9) + \mathrm{i}\,(3 - 2)$$
$$= 17 + i$$

$$z_1 + z_2 = (8 + 3i) - (9 - 2i)$$
$$= (8 - 9) + \mathrm{i}\,(3 + 2)$$
$$= -1 + 5i$$

$$z_1 z_2 = (8 + 3i)(9 - 2i)$$
$$= (72 - 16i + 27i - 6i^2)$$
$$= 78 + 11i$$

$$\frac{z_1}{z_2} = \frac{(8 + 3i)}{(9 - 2i)} = \frac{(8 + 3i)(9 + 2i)}{(9 - 2i)(9 + 2i)}$$

$$= \frac{(72 + 16i + 27i + 6i^2)}{81 - 4i^2}$$

$$= \frac{66}{85} + i\frac{43}{85}$$

Addition and Multiplication – Properties

1. Commutative law for addition

$$z_1 + z_2 = z_2 + z_1$$

2. Associative law for addition

$$z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$$

3. Additive identity

$$z + w = z, \qquad \forall z$$

$w = 0 + 0i$

4. Additive inverse

$$z + \delta = 0 + 0i$$

δ is additive inverse of z

5. Commutative law for multiplication

$$z_1 z_2 = z_2 z_1$$

6. Associative law for multiplication

$$z_1(z_2 z_3) = (z_1 z_2) z_3$$

7. Multiplicative identity

$$z\xi = z, \qquad \forall z$$

$\xi = 1 + 0i$

8. Multiplicative inverse

$$zz^{-1} = 1 + 0i$$

z^{-1} is multiplicative inverse of z

9. The Distributive law

$$z_1(z_2+z_3) = z_1z_2 + z_1z_3$$

Complex Conjugate

The conjugate of a complex number, z is denoted by $\bar{z}$, and given by

$$\bar{z} = x - iy$$

$$z + \bar{z} = (x + iy) + (x - iy) = 2x = \mathrm{Re}(z)$$

$$\therefore \mathrm{Re}(z) = \frac{1}{2}(z + \bar{z})$$

$$\mathrm{Im}(z) = \frac{1}{2i}(z - \bar{z})$$

Also,

$$\overline{z_1 + z_2} = \bar{z_1} + \bar{z_2}$$

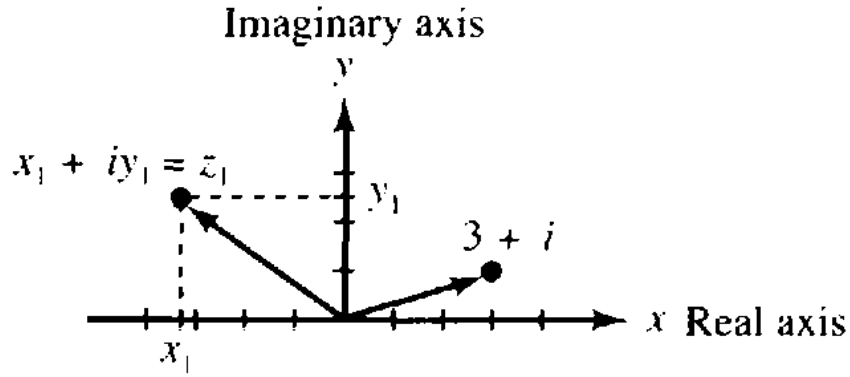
$$\overline{z_1 - z_2} = \bar{z_1} - \bar{z_2}$$

$$\overline{z_1 z_2} = \bar{z_1}\bar{z_2}$$

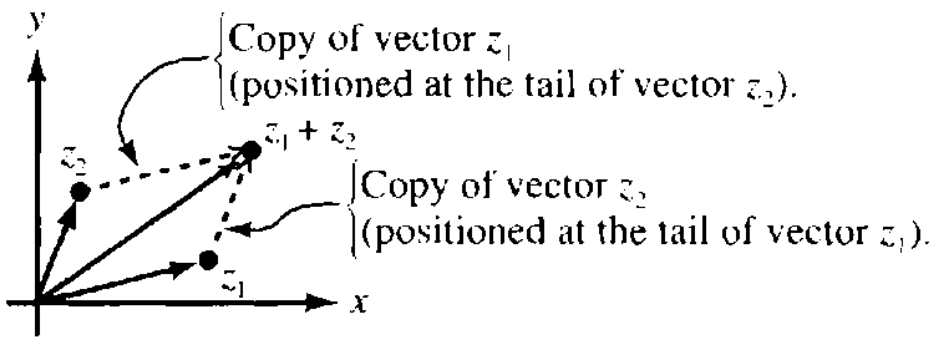
$$\frac{\bar{z_1}}{z_2} = \frac{\bar{z_1}}{\bar{z_2}}$$

The Geometry of Complex Numbers

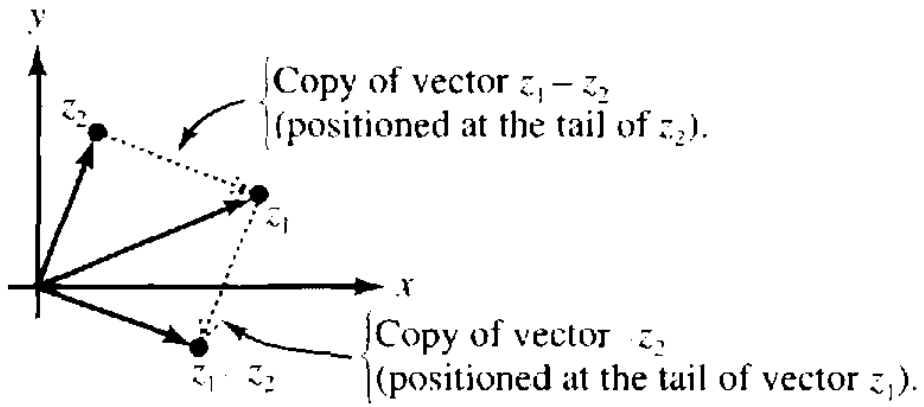
The $x - y$ plane in which a complex number is represented is called the complex plane.



The addition and subtraction of two complex numbers, z_1, z_2 can be obtained using parallelogram law

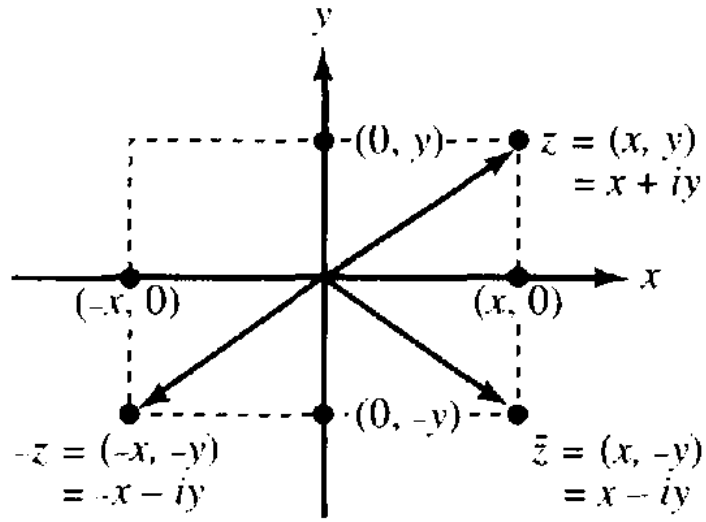


Addition of two complex numbers



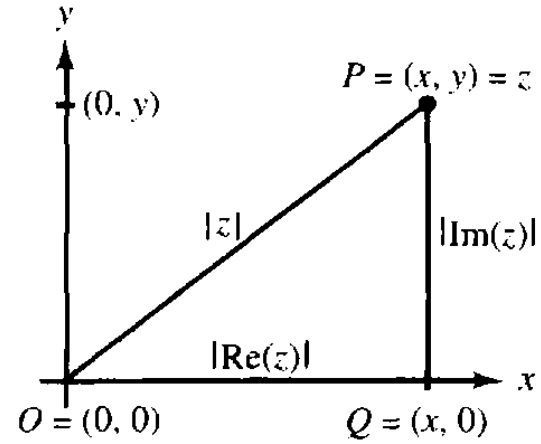
Subtraction of two complex numbers

The conjugate of a complex numbers, z is obtained geometrically by reflecting the point z in real – axis.



The modulus, or absolute value of a complex number, $z = x + iy$ is a non – negative real number

$$|z| = \sqrt{x^2 + y^2}$$



Also, from OPQ it can be seen that

$|x| \leq |z|, \qquad |y| \leq |z|$

$|\mathrm{Re}(z)| \leq |z|, \qquad |\mathrm{Im}(z)| \leq |z|$

$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = |z|^2$

$\therefore |z|^2 = z\bar{z}$

The Triangle Inequality

$$|z_1 + z_2| \le |z_1| + |z_2|$$

Proof:

$$|z_1 + z_2|^2 = (z_1 + z_2)\,\overline{(z_1 + z_2)}$$

$$= (z_1 + z_2)\,(\bar{z}_1 + \bar{z}_2)$$

$$= z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2$$

$$= |z_1|^2 + z_1\bar{z}_2 + \overline{(z_1\bar{z}_2)} + |z_2|^2$$

$$= |z_1|^2 + 2\,\mathrm{Re}(z_1\bar{z}_2) + |z_2|^2$$

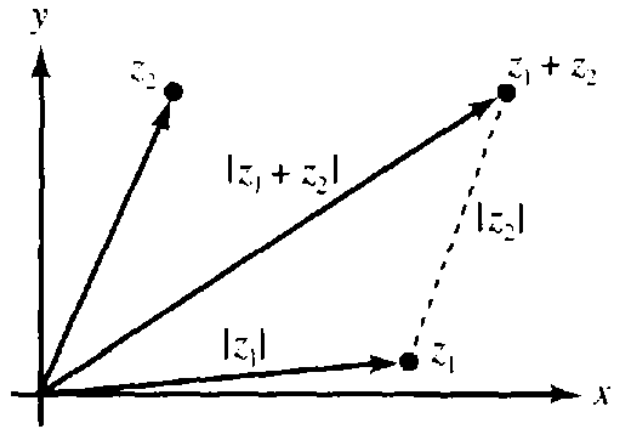
$$\le |z_1|^2 + 2|\mathrm{Re}(z_1\bar{z}_2)| + |z_2|^2$$

$$\le |z_1|^2 + 2|z_1\bar{z}_2| + |z_2|^2$$

$$\le |z_1|^2 + 2|z_1||\bar{z}_2| + |z_2|^2$$

$$\le |z_1|^2 + 2|z_1||z_2| + |z_2|^2$$

$$\le (|z_1| + |z_2|)^2$$



Geometrical representation of the triangle inequality

Taking square roots

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

By induction, the generalized triangle inequality is

$$|z_1 + z_2 + \ldots + z_n| \leq |z_1| + |z_2| + \ldots + |z_n|$$

Also,

$$|z_1 - z_2| \geq |z_1| - |z_2|$$

$$|z_1 z_2| = |z_1|\,|z_2|$$

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$$

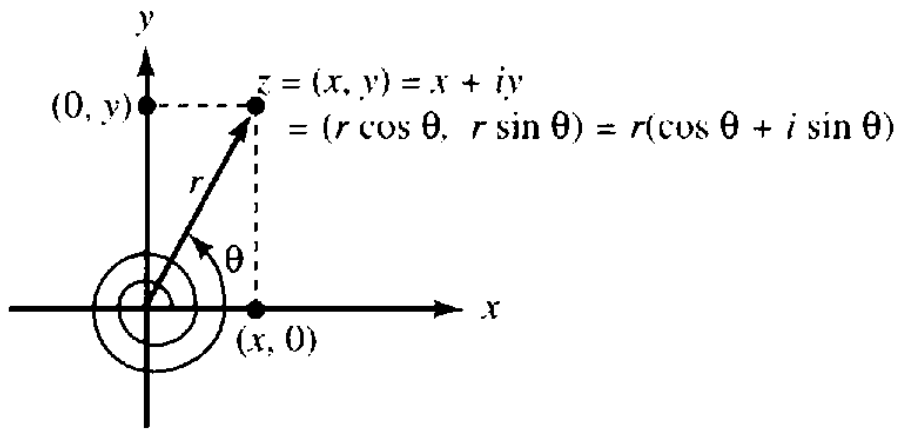
Polar Form of Complex Numbers

Using polar co – ordinates, r and θ

$$z = x + iy = r\cos\theta + i\, r\sin\theta$$

$$= r(\cos\theta + i\sin\theta)$$

$$r = |z| = \sqrt{x^2 + y^2}$$



θ is called the argument of z, denoted by $\arg(z)$

$$\theta = \arg(z) = \tan^{-1}\frac{y}{x}$$

- Since sine and cosine are periodic function with period 2π, one wants to specify a unique value of $\arg(z)$ if $z \neq 0$

- Principal value, $\mathrm{Arg}(z)$ of $\arg(z)$

$$-\pi < \mathrm{Arg}(z) \leq \pi$$

$\arg(z) = \mathrm{Arg}(z) + 2\pi k$, where k is an integer

Example

Find polar representation of $z_1 = \sqrt{3} + i$, $z_2 = -\sqrt{3} - i$

$$r = |z_1| = \sqrt{\sqrt{3}^2 + 1^2} = 2$$

$$\theta = \arg(z_1) = \tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

$$z_1 = r(\cos\theta + i\sin\theta) = 2(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$$

$$= 2\left[\cos\left(\frac{\pi}{6} + 2\pi n\right) + i\sin\left(\frac{\pi}{6} + 2\pi n\right)\right]$$

$$r = |z_2| = \sqrt{(-\sqrt{3})^2 + 1^2} = 2$$

$$\theta = \arg(z_2) = \tan^{-1}\frac{-1}{-\sqrt{3}} = \frac{7\pi}{6}$$

$$z_2 = r(\cos\theta + i\sin\theta) = 2(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6})$$

$$= 2\left[\cos\left(\frac{7\pi}{6} + 2\pi n\right) + i\sin\left(\frac{7\pi}{6} + 2\pi n\right)\right]$$

Multiplication and Division in Polar Form

$z_1 = r_1 (\cos\theta_1 + i \sin\theta_1) \qquad z_2 = r_2 (\cos\theta_2 + i \sin\theta_2)$

$z_1 z_2 = r_1 (\cos\theta_1 + i \sin\theta_1).r_2 (\cos\theta_2 + i \sin\theta_2)$

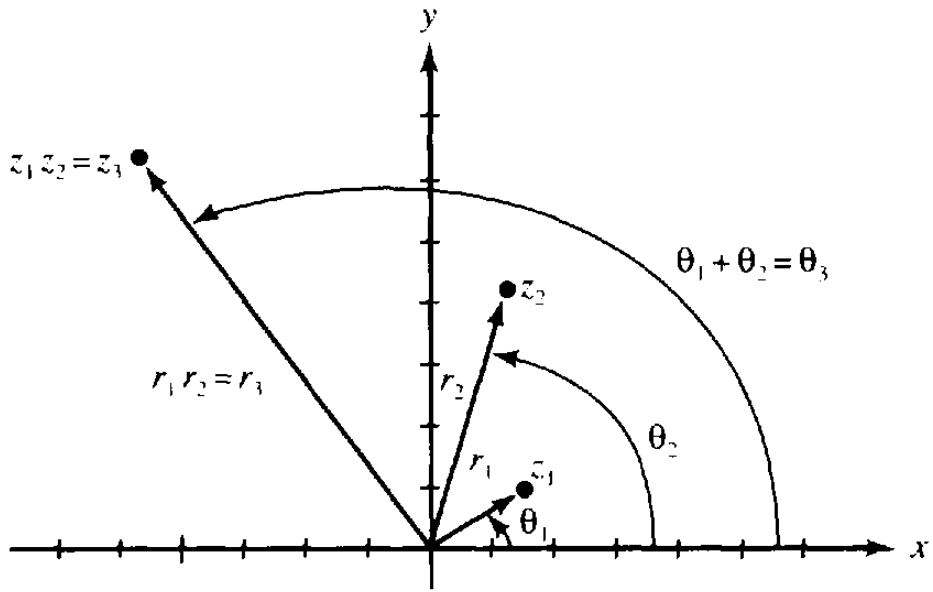
$= r_1 r_2[(\cos\theta_1 \cos\theta_2 + i \cos\theta_1 \sin\theta_2 + i \sin\theta_1 \cos\theta_2 + i^2 \sin\theta_1 \sin\theta_2]$

$= r_1 r_2[(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2) + i (\cos\theta_1 \sin\theta_2 + i \sin\theta_1 \cos\theta_2)]$

$= r_1 r_2[(\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$

$|z_1 z_2| = r_1 r_2 = |z_1|\,|z_2|$

$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$



$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\left[(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)\right.$$

$$\left|\frac{z_1}{z_2}\right| = \frac{r_1}{r_2} = \frac{|z_1|}{|z_2|}$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

Example

$z_1 = -2 + 2i,\ \ z_2 = 3i$

$|z_1| = 2\sqrt{2} \qquad \arg(z_1) = \frac{3\pi}{4}$

$|z_2| = 3 \qquad \arg(z_2) = \frac{\pi}{2}$

$z_1 z_2 = (-2 + 2i)(3i) = -6 - 6i$

$|z_1 z_2| = 6\sqrt{2} = |z_1|\,|z_2|$

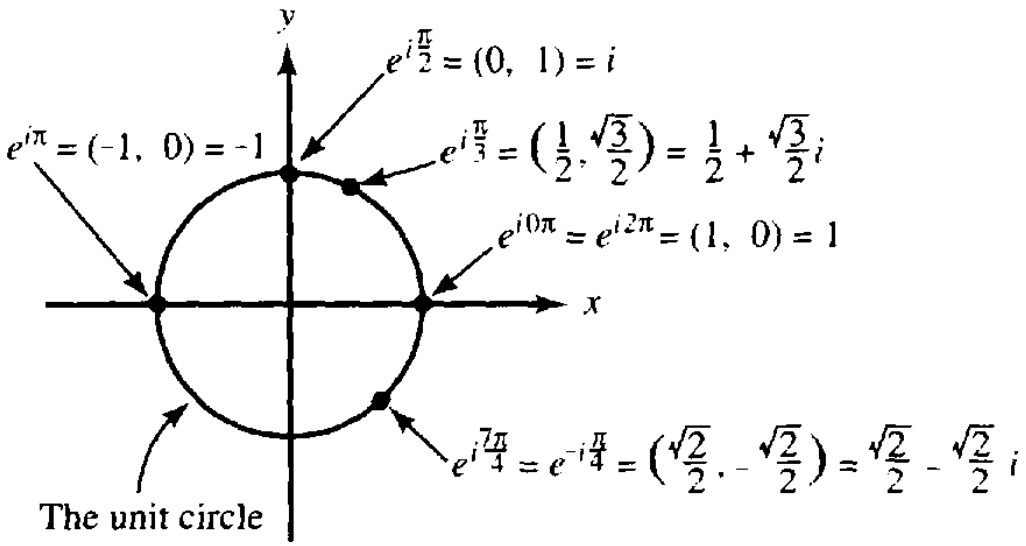
$$\arg(z_1 z_2) = \frac{5\pi}{4} = \frac{3\pi}{4} + \frac{\pi}{2} = \arg(z_1) + \arg(z_2)$$

Euler's Formula

$$e^{i\theta} = \cos\theta + i\ \sin\theta$$

If θ is any real number, $e^{i\theta}$ will be located on unit circle with radius 1 and centered at origin

$$\therefore \left|e^{i\theta}\right| = \sqrt{(\cos\theta)^2 + (\sin\theta)^2} = 1$$



Using this notation, any complex number $z = r(\cos\theta + i\ \sin\theta)$ can be written as:

$$z = re^{i\theta}$$

$$z_1 z_2 = (r_1 e^{i\theta_1})(r_2 e^{i\theta_2})$$

$$= r_1 r_2 e^{i(\theta_1+\theta_2)}$$

$$= r_1\, r_2[(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$$

$$z^{-1} = \frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r}e^{-i\theta}$$

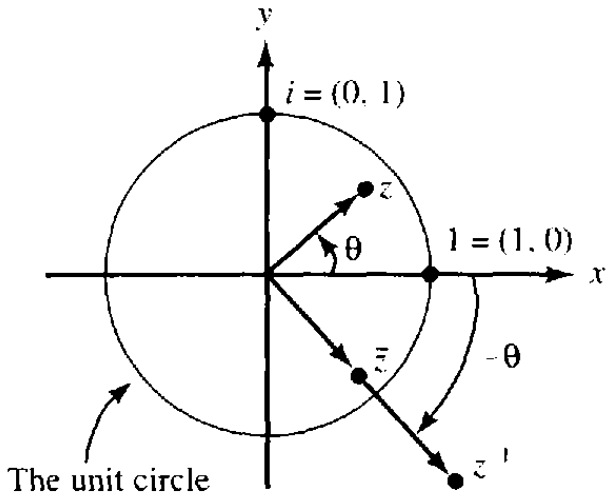
$$\bar{z} = r(\cos\theta - i\ \sin\theta)$$

$$= r(\cos(-\theta) + i\ \sin(-\theta))$$

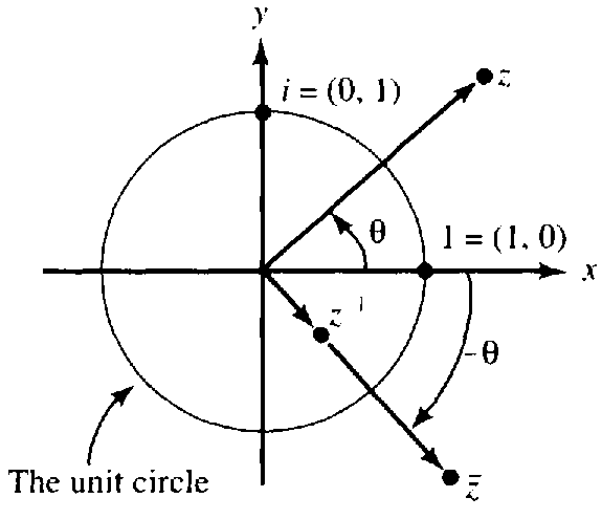
$$= re^{-i\theta}$$

$$z^n = \left(re^{i\theta}\right)^n = r^n e^{in\theta} = r^n(\cos n\theta + i\ \sin n\theta)$$

$$z^{-n} = \left(re^{i\theta}\right)^{-n} = r^{-n}e^{-in\theta} = r^{-n}(\cos n\theta - i\ \sin n\theta)$$



$|z| < 1$



$|z| > 1$

Also,

$$\left(e^{i\theta}\right)^n = (\cos n\theta + i \sin n\theta)$$

$$\therefore (\cos\theta + i \sin\theta)^n = (\cos n\theta + i \sin n\theta) \qquad \text{(De – Moivre's Formula)}$$

Example

Find $\cos 3\theta,\ \sin 3\theta$ using De Moivre's formula

$$(\cos\theta + i \sin\theta)^3 = \cos^3\theta + i^3\sin^3\theta + 3i\cos\theta\sin\theta\,(\cos\theta + i \sin\theta)$$

$$= (\cos^3\theta - 3\cos\theta\sin^2\theta) + i(-\sin^3\theta + 3\cos^2\theta\sin\theta)$$

$$= \cos 3\theta + i \sin 3\theta$$

$$\therefore \cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$$

$$= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta)$$

$$= 4\cos^3\theta - 3\cos\theta$$

$$\sin 3\theta = -\sin^3\theta + 3\cos^2\theta\sin\theta$$

$$= -\sin^3\theta + 3\sin\theta(1 - \sin^2\theta)$$

$$= 3\sin\theta - 4\sin^3\theta$$

Roots of Complex Numbers

Consider,

$$z = w^n$$

For a given $z \neq 0$, there corresponds precisely n distinct values of w; each of these is called the nth root of z, and we write

$$w = \sqrt[n]{z}$$

Let us try to find n – values of $\sqrt[n]{z}$

$$z = r(\cos\theta + i\ \sin\theta) = re^{i\theta}$$

$$w = \rho e^{i\phi}$$

$$w^n = z$$

$$\Rightarrow \left(\rho e^{i\phi}\right)^n = re^{i\theta}$$

$$\rho^n e^{in\phi} = re^{i\theta}$$

$$\therefore\ \rho^n\ (\cos n\phi + i\ \sin n\phi) = r\ (\cos\theta + i\ \sin\theta)$$

Equating modulus and arguments

$$\rho = \sqrt[n]{r}$$

$$n\phi = \theta + 2k\pi, \qquad k = 0,1,2, \dots (n-1)$$

$$\phi = \frac{\theta}{n} + \frac{2k\pi}{n}, \qquad k = 0,1,2, \dots (n-1)$$

Therefore, $\sqrt[n]{z}$ has n – values

$$w = \sqrt[n]{z} = \sqrt[n]{r}\left(\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right) \qquad k = 0,1,2, \dots (n-1)$$

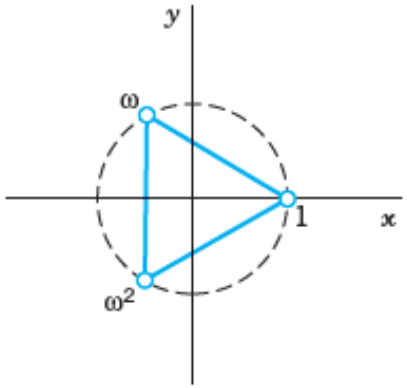
Now, if we take $z = 1$, then $r = 1$ and $\theta = \mathrm{Arg}(z) = 0$

$$\sqrt[n]{1} = \left(\cos\left(\frac{2k\pi}{n}\right) + i\sin\left(\frac{2k\pi}{n}\right)\right) \qquad k = 0,1,2, \dots (n-1)$$

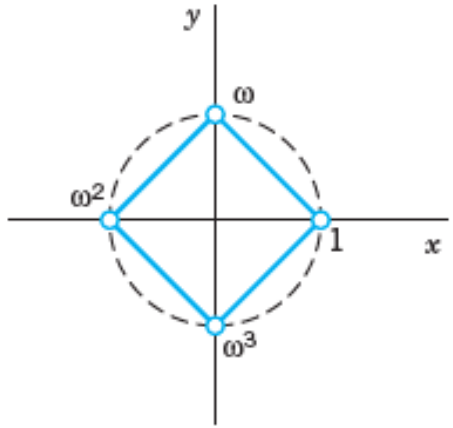
These, n – values lie on a unit circle.

$$\sqrt[3]{1} = 1, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

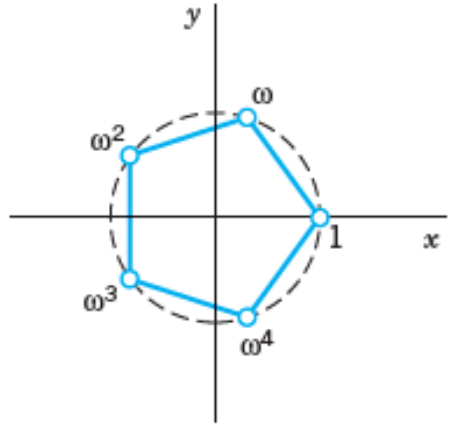
$$\sqrt[4]{1} = \pm 1, \pm i$$



$\sqrt[3]{1}$



$\sqrt[4]{1}$



$\sqrt[5]{1}$

If ω denotes the value corresponding to $k = 1$ then $\sqrt[n]{1}$ can be written as

$$1, \omega, \omega^2, \ldots, \omega^{n-1}$$

More generally, if w_1 is any nth root of an arbitrary complex number $z \neq 0$, then $\sqrt[n]{z}$ can be written as

$$w_1, w_1\omega, \ w_1\omega^2 \ , \ldots, w_1\omega^{n-1}$$

The Topology of Complex Numbers

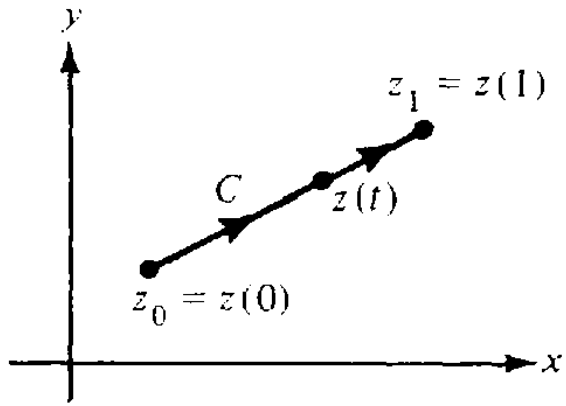
We define a curve C in a complex plane as

$$C: z(t) = x(t) + iy(t), \qquad a \le t \le b$$

$x(t),\ y(t)$ are continuous real valued function

If both $x(t),\ y(t)$ are differentiable functions, the curve is smooth

Example: If $z_0 = x_0 + iy_0$, $z_1 = x_1 + iy_1$, then the straight-line segment joining z_0 to z_1



$$C: z(t) = z_0 + (z_1 - z_0)t, \qquad 0 \le t \le 1$$

$$-C: w(t) = z_1 + (z_0 - z_1)t, \qquad 0 \le t \le 1$$

Also,

$$w(t) = z(1 - t)$$

$$-C: z(1 - t), \qquad 0 \le t \le 1$$

- A curve C with property, $z(a) = z(b)$ is said to be a closed curved

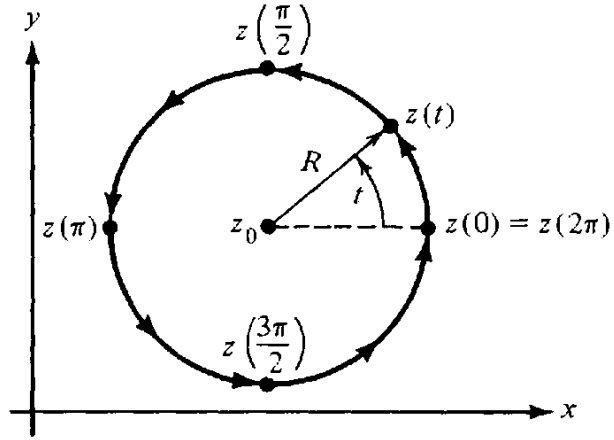
- A curve C is called simple if it does not cross over itself, which is expressed as,

$$z(t_1) \neq z(t_2) \text{ if } t_1 \neq t_2$$

except, possibly when $t_1 = a,\ t_2 = b$ (a simple closed curve)

- A circle with center $z_0 = x_0 + iy_0$ and radius R can be parametrized to form a simple closed curve

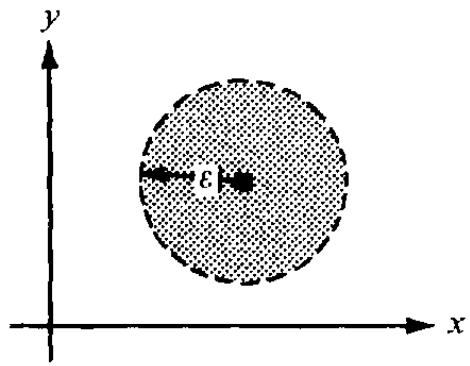
$$C:\ z(t) = z_0 + Re^{it}, \qquad 0 \leq t \leq 2\pi$$



- Consider the set of all points satisfying the inequality

$$|z_0 - z_1| < \epsilon$$

The set is an open circular disk of radius $\epsilon > 0$ about z_0, and is known as $\epsilon -$ neighborhood of z_0



- open disk of radius $\in$ centered at z_0

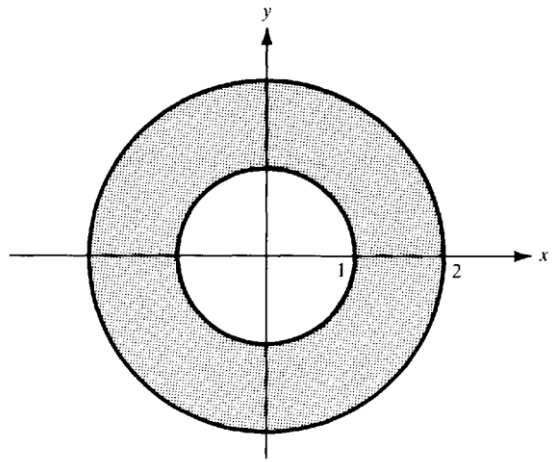
$$\mathrm{D}_{\in}(z_0) = \{z: \ |z - z_0| < \in \}$$

- closed disk of radius $\in$ centered at z_0

$$\overline{\mathrm{D}}_{\in}(z_0) = \{z: \ |z - z_0| \leq \in \}$$

- Annulus centered at z_0

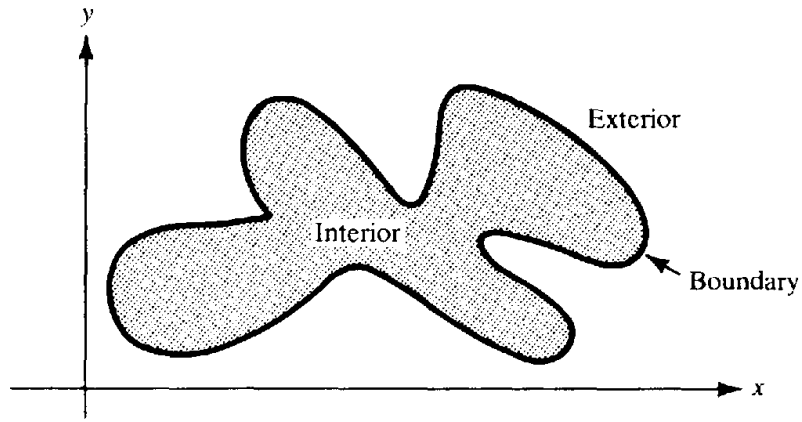
$$\mathrm{A}(z_0) = \{z: \in_1 < \ |z - z_0| < \in_2 \}$$



- Interior point: The point z_0 is said to be an interior point of set S provided there exists a $\in$ – neighborhood of z_0 that contains only point in S

$$\mathrm{D}_{\in}(z_0) \subseteq \mathrm{S}, \qquad \in > 0$$

- Exterior point: The point z_0 is said to be an exterior point of set S provided there exists a $\in$ – neighborhood of z_0 that contains no point of S



- Boundary point: If point z_0 is neither an interior point nor exterior point of S, then it is called a boundary point.

$$\mathrm{D}_{\in}(z_0) \cap \mathrm{S} \neq \phi \text{ and } \mathrm{D}_{\in}(z_0) \cap \mathrm{S}^c \neq \phi, \qquad \in > 0$$

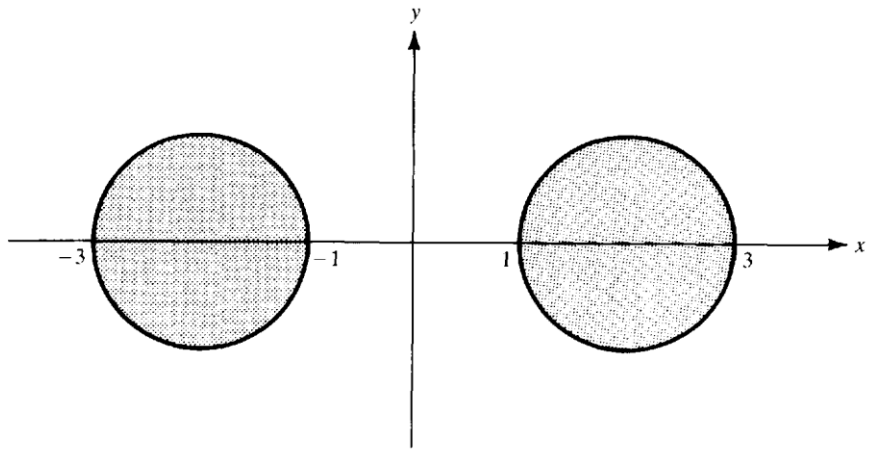
- A set is called open if every point of set S is an interior point of S

- A set S is called closed if it contains all its boundaries

- A set S is called connected if every pair of points z_1 and z_2 can be joined by a curve that lies entirely in S

$D(0) = \{z: \ |z| < 1\}$ is an open connected set

$B = \{z: \ |z+2| < 1 \text{ or } |z-2| < 1\}$ consists of two disjoint disk, hence it is not connected



Domain: open connected sets are known as domains

Functions of a Complex Variable

A function f of a complex variable z is a rule that assigns to each value of z in a set D (domain of definition of f) one and only one complex value, w

$$w = f(z)$$

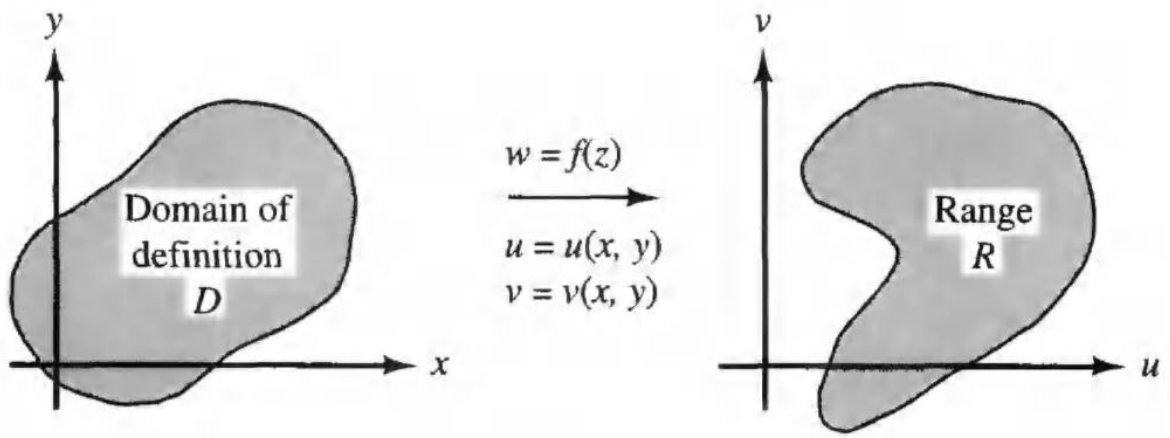
Range of f

$$\mathrm{R} = \{w = f(z):\ z \in \mathrm{D}\}$$

$z = x + iy$, then $w = u + iv$

$$w = f(x + iy) = u(x, y) + iv(x, y)$$

$u(x, y), v(x, y)$ are real functions of real variables, x, y



Also, $z = re^{i\theta}$, then

$$f(z) = f\left(re^{i\theta}\right) = \ u(r,\theta) + iv(r,\theta)$$

Example

$w = f(z) = \ z^2 + 3z$. Find . Calculate $u(x,y), v(x,y)$ value of at $z = 1 + 3i$

$$f(z) = \ z^2 + 3z = \ (x+iy)^2 + 3(x+iy)$$

$$= \ x^2 + i^2y^2 + 2i\,xy + 3x + 3iy$$

$$= \ (x^2 - y^2 + 3x) + i(2\,xy + 3y)$$

$$f(z) = \ u(x,y) + iv(x,y)$$

$$u(x,y) = x^2 - y^2 + 3x$$

$$v(x,y) = 2xy + 3y$$

$$f(1+3i) = \ (1 - 9 + 3) + i(6 + 9)$$

$$= -5 + 15i$$

Let $f(z)$ be a complex function of variable z that is defined for all values of z in some neighborhood of z_0. We say that f is continuous at z_0 if the following three conditions are met:

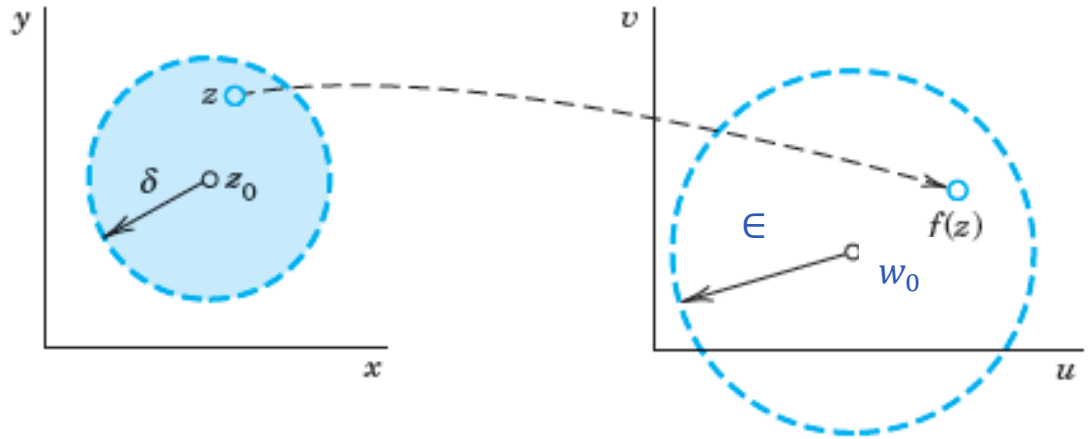
- $\lim_{z \to z_0} f(z)$ exists
- $f(z_0)$ exists
- $\lim_{z \to z_0} f(z) = f(z_0)$

Let $f(z)$ be a complex function of variable z. We say that f has a limit w_0 as z approaches z_0 , provided that the value of f gets close to value w_0 as z gets close to z_0. Mathematically,

$$\lim_{z \to z_0} f(z) = w_0$$

More precisely, for each positive number $\in > 0$, there exists a $\delta > 0$ such that

$$|f(z) - w_0| < \in \text{ whenever } 0 < |z - z_0| < \delta$$



Differentiable Functions

Let $f(z)$ be a complex function of variable z that is defined for all values of z in some neighborhood of z_0. Then,

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

Provided that the limit exists

Alternatively, $\Delta z = z - z_0$

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

If we let, $w = f(z)$ and $\Delta w = f(z) - f(z_0)$

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \frac{dw}{dz}$$

Example

$w = f(z) = z^3$. Find $f'(z)$

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{(z + \Delta z)^3 - (z)^3}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{(z + \Delta z - z)(2z\Delta z + \Delta z^2 + z^2 + z\Delta z + z^2)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} 3z^2 + 3z\Delta z + \Delta z^2$$

$$f'(z) = 3z^2$$

Differentiation rules for complex functions:

$$- \frac{d}{dz}(c + id) = 0$$

$$- \frac{d}{dz}(z^n) = nz^{n-1}$$

$$- \frac{d}{dz}\big((c + id)f(z)\big) = (c + id)f'(z)$$

$$- \frac{d}{dz}\big(f(z) + g(z)\big) = f'(z) + g'(z)$$

$$- \frac{d}{dz}\big(f(z)g(z)\big) = f'(z)g(z) + f(z)g'(z)$$

$$-\frac{d}{dz}\left(\frac{f(z)}{g(z)}\right) = \frac{f'(z)g(z) - f(z)g'(z)}{g(z)^2}$$

$$-\frac{d}{dz}\left(\frac{1}{z^n}\right) = \frac{-n}{z^{n+1}}, \qquad z \neq 0 \text{ and n is positive integer}$$

Example

$w = f(z) = z^2 + 2iz + 3$. Find $f'(z)$

$$f'(z) = \frac{d}{dz}(z^2 + 2iz + 3)$$

$$= \frac{d}{dz}(z^2) + \frac{d}{dz}(2iz) + \frac{d}{dz}(3)$$

$$= 2(z + i)$$

Example

$w = f(z) = (z^2+2iz + 3)^4$. Find $f'(z)$

$$f'(z) = \frac{d}{dz}(z^2+2iz + 3)^4 = 4(z^2+2iz + 3)^4.2(z + i)$$

$$= 8(z^2+2iz + 3)^3(z + i)$$

Analytic Functions

A function $f(z)$ is said to be analytic in a domain D if $f(z)$ is defined and differentiable at all points of D

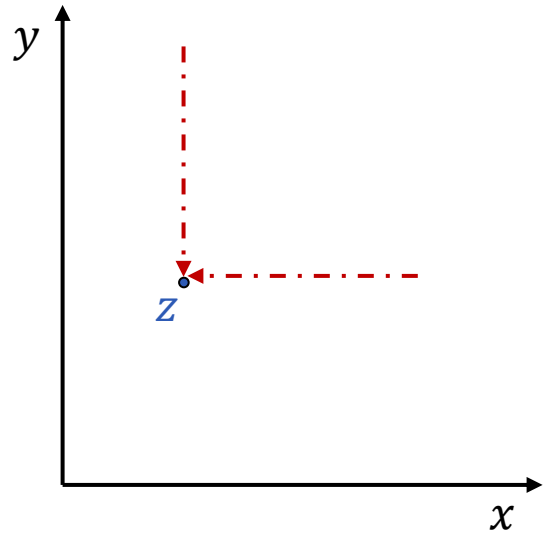
- A function $f(z)$ is analytic at a point $\mathrm{z} = z_0$ in domain D if $f(z)$ is analytic in a neighborhood of z_0

Example of Analytic functions:

- $1, z, z^2$
- $f(z) = \mathrm{C}_0 + \mathrm{C}_1 z + \mathrm{C}_2 z^2 + \ldots + \mathrm{C}_n z^n$
- $f(z) = \dfrac{g(z)}{h(z)}$ is analytic except at points $h(z) = 0$

Cauchy – Riemann Equations

Consider the function $f(z) = \bar{z} = x - iy$



$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{\overline{(z + \Delta z)} - \bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y}$$

First, we approach z along a line parallel to x – axis, then $\Delta y = 0$

$$\lim_{\Delta z \to 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = 1$$

First, we approach z along a line parallel to y – axis, then $\Delta x = 0$

$$\lim_{\Delta z \to 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = -1$$

Since, the limits along the two – directions are different, therefore $f(z) = \bar{z}$ is not differentiable.

Consider the complex function $f(z) = u(x,y) + iv(x,y)$

f is analytic in a domain D if and only if the first partial derivatives of $u(x,y),\ v(x,y)$ satisfy the Cauchy – Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Conversely, let $f(z) = u(x,y) + iv(x,y)$ be defined and continuous in some neighborhood of a point, and differentiable at itself. Then, at that point, the first partial derivatives of $u\ v$ exists, and satisfy the Cauchy – Riemann equations.

Also, if is analytic in a domain D, these partial derivatives exists and satisfy the Cauchy – Riemann equations at all points.

Example

$w = f(z) = \bar{z} = x - iy$

$u = x, \qquad v = -y$

$u_x = 1, \qquad u_y = 0 \qquad v_x = 0, \qquad v_y = -1$

$u_x \neq v_y, \qquad u_y = -v_x = 0$

$f(z)$ is not differentiable.

Example

$w = f(z) = e^z = e^{x+iy} = e^x(\cos y + i \sin y)$

$u = e^x \cos y, \qquad v = e^x \sin y$

$u_x = e^x \cos y, \qquad u_y = -e^x \sin y$

$v_x = e^x \sin y, \qquad v_y = e^x \cos y$

$\therefore\ u_x = v_y, \qquad u_y = -v_x$

$f(z)$ is analytic for all z

Remarks:

$-f'(z) = u_x(x,y) + i\, v_x(x,y)$

$-f'(z) = v_y(x,y) - i\ u_y(x,y)$

Example

$w = f(z) = e^{-y}(\cos x + i \sin x)$. Calculate $f'(z)$

$u = e^{-y} \cos x, \qquad v = e^{-y} \sin x$

$$u_x = -e^{-y}\sin x\,, \qquad u_y = -\,e^{-y}\cos x$$

$$v_x = e^{-y}\cos x\,, \qquad v_y = -e^{-y}\sin x$$

$$\therefore\ u_x = v_y, \qquad u_y = -v_x$$

$f(z)$ is analytic for all z

$$f'(z) = u_x(x,y) + i\,v_x(x,y)$$
$$= e^{-y}(-\sin x + i\cos x)$$

If we use polar form of $z = r(\cos\theta + i\,\sin\theta)$, and set

$$f(z) = u(r,\theta) + iv(r,\theta)$$

Cauchy – Riemann equations are:

$$u_r = \frac{1}{r}v_\theta, \qquad u_\theta = -rv_r$$

$$-f'(z) = e^{-i\theta}[u_r(r,\theta) + i\,v_r(r,\theta)]$$

$$-f'(z) = \frac{1}{r}e^{-i\theta}[v_\theta(r,\theta) - i\,u_\theta(r,\theta)]$$

Example

$$w = f(z) = z^{\frac{1}{2}} = r^{\frac{1}{2}}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$$

$$u = r^{\frac{1}{2}}\cos\frac{\theta}{2}, \qquad v = r^{\frac{1}{2}}\sin\frac{\theta}{2}$$

$$u_r = \frac{1}{2}r^{-\frac{1}{2}}\cos\frac{\theta}{2}, \qquad u_\theta = -\frac{1}{2}r^{\frac{1}{2}}\sin\frac{\theta}{2}$$

$$v_r = \frac{1}{2}r^{-\frac{1}{2}}\sin\frac{\theta}{2}, \qquad v_\theta = \frac{1}{2}r^{\frac{1}{2}}\cos\frac{\theta}{2}$$

$$\therefore\ u_r = \frac{1}{r}v_\theta, \qquad u_\theta = -rv_r$$

$f(z)$ is analytic for all z

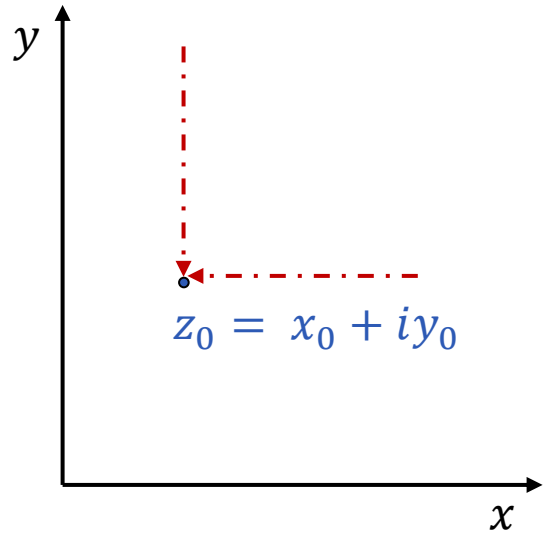
$$f'(z) = e^{-i\theta}[u_r(r,\theta) + i\,v_r(r,\theta)]$$

$$= e^{-i\theta}\left[\frac{1}{2}r^{-\frac{1}{2}}\cos\frac{\theta}{2} + i\frac{1}{2}r^{-\frac{1}{2}}\sin\frac{\theta}{2}\right]$$

$$= e^{-i\theta}\left(\frac{1}{2}r^{-\frac{1}{2}}e^{\frac{i\theta}{2}}\right) = \frac{1}{2}r^{-\frac{1}{2}}e^{-\frac{i\theta}{2}} = \frac{1}{2z^{\frac{1}{2}}}$$

Cauchy – Riemann Equations - Proof

Consider the complex function $f(z)$ and we chose horizontal and vertical lines that pass-through point $(x_0,\ y_0)$, and compute the derivative.



$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

For horizontal approach to z_0, we set $z = x + iy_0$

$$f'(z_0) = \lim_{(x,y_0)\to(x_0,y_0)} \frac{f(x + iy_0) - f(x_0 + iy_0)}{x + iy_0 - (x_0 + iy_0)}$$

$$= \lim_{(x,y_0)\to(x_0,y_0)} \frac{[u(x,\ y_0) + iv(x,\ y_0)] - [u(x_0,\ y_0) + iv(x_0,\ y_0)]}{x\ -\ x_0}$$

$$= \lim_{x\to x_0} \frac{u(x,\ y_0) - u(x_0,\ y_0)}{x\ -\ x_0} + i \lim_{x\to x_0} \frac{v(x,\ y_0) - v(x_0,\ y_0)}{x\ -\ x_0}$$

$$f'(z_0) = u_x + iv_x \qquad (1)$$

For vertical approach to z_0, we set $z = x_0 + iy$

$$f'(z_0) = \lim_{(x,y_0)\to(x_0,y_0)} \frac{f(x_0+iy) - f(x_0+iy_0)}{x_0+iy-(x_0+iy_0)}$$

$$= \lim_{(x,y_0)\to(x_0,y_0)} \frac{[u(x_0,\ y)+iv(x,\ y_0)]-[u(x_0,\ y)+iv(x_0,\ y_0)]}{i(y\ -\ y_0)}$$

$$= \lim_{y\to y_0} \frac{u(x_0,\ y)-u(x_0,\ y_0)}{i(y\ -\ y_0)} + i \lim_{y\to y_0} \frac{v(x_0,\ y)-v(x_0,\ y_0)}{i(y\ -\ y_0)}$$

$$f'(z_0) = -iu_y + v_y \qquad (2)$$

Since, f is differentiable at z_0, the limits given by (1) and (2) must be equal

$$u_x + iv_x = v_y - iu_y$$

$$u_x = v_y, \qquad v_x = -u_y$$

Elementary Functions - Exponential

The complex exponential function is defined as:

$$e^z = \exp z = \sum_{n=1}^{\infty} \frac{z^n}{n!}$$

The function e^z is an entire function satisfying:

- $(e^z)' = \frac{d}{dz} e^z = e^z$
- $e^{z_1+z_2} = e^{z_1} . e^{z_2}$
- $e^{i\theta} = \cos\theta + i \sin\theta$, θ is a real number

$$e^z = e^{x+iy} = e^x (\cos y + i \sin y)$$

$u = e^x \cos y$, $v = e^x \sin y$

$u_x = e^x \cos y$, $u_y = -e^x \sin y$

$v_x = e^x \sin y$, $v_y = e^x \cos y$

$\therefore u_x = v_y$, $u_y = -v_x$

Cauchy – Riemann conditions are satisfied for all z

$$(e^z)' = u_x(x,y) + i\,v_x(x,y)$$
$$= e^x(\cos y + i \sin y)$$
$$= e^z$$

Let us consider two complex numbers, $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$

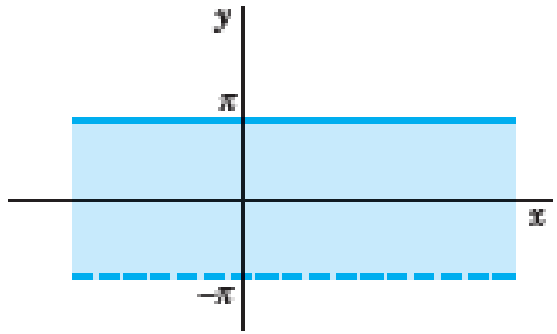
$$e^{z_1}.e^{z_2} = e^{x_1}(\cos y_1 + i \sin y_1)e^{x_2}(\cos y_2 + i \sin y_2)$$
$$= e^{x_1+x_2}(\cos(y_1+y_2) + i \sin(y_1+y_2))$$
$$= e^{z_1+z_2}$$

Periodicity of e^z with period $2\pi i$

$$e^{z+2\pi i} = e^z \text{ for all } z$$

Hence, all values that $w = e^z$ can assume are already assumed in the horizontal strip of width 2π

Hence, This strip is called fundamental region of e^z.



- $e^{z_1} = e^{z_2}$ iff $z_2 = z_2 + 2\pi ni$, n is an integer
- $e^{z} = 1$ iff $z = 2\pi ni$, n is an integer

Elementary Functions - Logarithmic

$z \neq 0$, we define the function as the inverse of exponential function:

$$\log z = w \text{ iff } z = e^{iw}$$

$$\log z = \ln|z| + i \arg z$$

$$\log z = \ln|z| + i\,(\operatorname{Arg} z + 2\pi n), \qquad n \text{ is an integer}$$

For $z \neq 0$, the principal value of logarithm is defined as:

$$\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z \qquad -\pi < \operatorname{Arg}(z) \leq \pi$$

Example

$$\log(1+i) = \ln|1+i| + i \arg(1+i)$$

$$= \ln\sqrt{2} + i\left(\frac{\pi}{4} + 2\pi n\right)$$

$$= \frac{1}{2}\ln 2 + i\left(\frac{\pi}{4} + 2\pi n\right)$$

$$\operatorname{Log} z = \frac{1}{2}\ln 2 + i\frac{\pi}{4}$$

Properties:

- $e^{\mathrm{Log}\, z} = z \qquad \forall\, z \neq 0$
- $\mathrm{Log}(x + i0) = \ln x \ \text{ when } x > 0$
- $z = re^{i\theta}$

$$\mathrm{Log}(z) = \ln r + i\theta \ \text{ where } r > 0, \qquad -\pi < \mathrm{Arg}(z) \leq \pi$$

The function $\log z$ obeys the familiar properties of logarithm:

- $\log(z_1 z_2) = \log(z_1) + \log(z_2)$
- $\log\left(\frac{z_1}{z_2}\right) = \log(z_1) - \log(z_2)$
- $\log\left(\frac{1}{z}\right) = -\log(z)$

Example

$z_1 = z_1 = e^{i\pi} = -1$

Taking principal values,

$$\mathrm{Log}(z_1) = \mathrm{Log}(z_2) = i\pi$$

$$\Rightarrow \mathrm{Log}(z_1 z_2) = \mathrm{Log}(1) = 0$$

$\therefore \log(z_1 z_2) \neq \mathrm{Log}(z_1) + \mathrm{Log}(z_2)$

$\log(z_1 z_2) = \log(1) = 2\pi i = \log(z_1) + \log(z_2)$

– $(\log z)' = \frac{1}{z}$, where $r > 0, \quad -\pi < \mathrm{Arg}(z) \leq \pi$

$u = \ln r, \quad v = \theta$

$v_r = 0, \quad v_\theta = 1$

$u_r = \frac{1}{r}, \quad u_\theta = 0$

$\therefore u_r = \frac{1}{r} v_\theta, \quad u_\theta = -r v_r$

$(\log z)' = e^{-i\theta}[u_r(r,\theta) + i\, v_r(r,\theta)]$

$= e^{-i\theta}.\frac{1}{r} = \frac{1}{z}$, where $r > 0, \quad -\pi < \mathrm{Arg}(z) \leq \pi$

In particular, $f(z) = \mathrm{Log}\ z$ is discontinuous at $z = 0$, and at points on the negative real – axis. Hence, the derivative of $\mathrm{Log}\ z$ does not exist at $z = 0$, and at points on negative real – axis.

Elementary Functions - Trigonometric

The trigonometric sine and cosine function are defined as:

$$\cos z = \sum_{n=1}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} \qquad \sin z = \sum_{n=1}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}$$

$$\therefore \cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Furthermore:

$$\tan z = \frac{\sin z}{\cos z}, \qquad \cot z = \frac{\cos z}{\sin z}$$

$$\sec z = \frac{1}{\cos z}, \qquad \operatorname{cosec} z = \frac{1}{\sin z}$$

Properties:

$$\cos(x + i0) = \cos x$$

$$\sin(x + i0) = \sin x$$

- $\sin z$ and $\cos z$ are entire functions
- $(\sin z)' = \left(\frac{e^{iz} - e^{-iz}}{2i}\right)' = \frac{1}{2i}(ie^{iz} + ie^{-iz})$

$$= \frac{1}{2}(e^{iz} + e^{-iz})$$

$$= \cos z$$

- $(\cos z)' = -\sin z$
- $\sin(-z) = -\sin z$ and $\cos(-z) = \cos z$
- $\sin^2 z + \cos^2 z = 1$
- $\sin(z_1 + z_2) = \sin z_1 \cos z_1 + \cos z_1 \sin z_2$

  $\cos(z_1 + z_2) = \cos z_1 \cos z_1 - \sin z_1 \sin z_2$
- $\sin(z + 2n\pi) = \sin z,$ n is an integer

  $\cos(z + 2n\pi) = \cos z,$ n is an integer

$$-\cos z = \frac{e^{iz} + e^{-iz}}{2} = \frac{e^{i(x+iy)} + e^{-i(x+iy)}}{2}$$

$$= \frac{e^{-y+xi} + e^{y-xi}}{2}$$

$$= \frac{1}{2}[e^{-y}(\cos x + i \sin x) + e^{y}(\cos x - i \sin x)\,]$$

$$= \cos x \left(\frac{e^{y} + e^{-y}}{2}\right) - i \sin x \left(\frac{e^{y} - e^{-y}}{2}\right)$$

$$= \cos x \cosh y - i \sin x \sinh y$$

$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

The hyperbolic sine and cosine functions are defined as:

$$\therefore \cosh z = \frac{e^z + e^{-z}}{2}, \qquad \sinh z = \frac{e^z - e^{-z}}{2}$$

Furthermore:

$$\tanh z = \frac{\sinh z}{\cosh z}, \qquad \coth z = \frac{\cosh z}{\sinh z}$$

$$\operatorname{sech} z = \frac{1}{\cosh z}, \qquad \operatorname{cosech} z = \frac{1}{\sinh z}$$

- $\sinh(-z) = -\sinh z$ and $\cosh(-z) = \cosh z$
- $\cosh^2 z - \sinh^2 z = 1$
- $\sinh(z_1 + z_2) = \sinh z_1 \cosh z_1 + \cosh z_1 \sinh z_2$

$$\cosh(z_1 + z_2) = \cosh z_1 \cosh z_1 + \sinh z_1 \sinh z_2$$

The trigonometric and hyperbolic functions are related by the following identities:

$$\cosh(iz) = \cos z, \qquad \sinh(iz) = i \sin z$$

$$\cos(iz) = \mathrm{i} \sinh z, \qquad \cos(iz) = \cosh z$$

Inverse trigonometric functions are:

$$-\ \sin^{-1}(z) = -i\log[iz + \ (1\ -\ z^2)^{\frac{1}{2}}]$$

$$-\ \cos^{-1}(z) = -i\log[z + i(1\ -\ z^2)^{\frac{1}{2}}]$$

$$-\ \tan^{-1}(z) = -\frac{i}{2}\log\left(\frac{i+z}{i\ -z}\right)$$

$$-\ \frac{d}{dz}(\sin^{-1}(z)) \ = \frac{1}{(1\ -\ z^2)^{\frac{1}{2}}}$$

$$-\ \frac{d}{dz}(\cos^{-1}(z)) \ = -\ \frac{1}{(1\ -\ z^2)^{\frac{1}{2}}}$$

$$-\ \frac{d}{dz}(\tan^{-1}(z)) \ = \frac{1}{1+\ z^2}$$

Example

$$\sin^{-1}\left(\sqrt{2}\ \right)\ =\ -i\log[i\sqrt{2} + \left(1\ -\ \sqrt{2}^2\right)^{\frac{1}{2}}]$$

$$=\ -i\log[i\sqrt{2} \pm i]$$

$$= -i\log[i(\sqrt{2} \pm 1)]$$

$$= i\left[\ln(\sqrt{2} \pm 1) + i\left(\frac{\pi}{2} + 2\pi n\right)\right]$$

$$= \left(\frac{\pi}{2} + 2\pi n\right) - i\ln(\sqrt{2} \pm 1)$$

Inverse hyperbolic functions are given by equations:

- $\sinh^{-1}(z) = \log[z + (1 + z^2)^{\frac{1}{2}}]$
- $\cosh^{-1}(z) = \log[z + (z^2 - 1)^{\frac{1}{2}}]$
- $\tanh^{-1}(z) = \frac{1}{2}\log\left(\frac{1+z}{1-z}\right)$
- $\frac{d}{dz}(\sin^{-1}(z)) = \frac{1}{(z^2+1)^{\frac{1}{2}}}$
- $\frac{d}{dz}(\cos^{-1}(z)) = \frac{1}{(z^2-1)^{\frac{1}{2}}}$
- $\frac{d}{dz}(\tan^{-1}(z)) = \frac{1}{1-z^2}$

Elementary Functions – Complex Exponential

Let c be a complex number. We define z^c by the equation

$$z^c = e^{c \log z}$$

Since, $\log z$ is a multi – valued function, the function z^c will be in general multi – valued. The principal branch of multi – valued function is given by:

$$z^c = e^{c \operatorname{Log} z}$$

Properties:

- $z^c = \dfrac{1}{z^c}$
- $z^c z^d = z^{c+d}$
- $(z^c)^n = z^{nc}$, n is an integer
- $\dfrac{d}{dz}(z^c) = \dfrac{c}{z} z^c$

Example: Find the principal value of $(1+i)^{\frac{1}{2}}$

$$\therefore\ z^c = e^{c\log z};\ c = \frac{1}{2},\qquad z = 1+i$$

$$\log(1+i) = \ln|1+i| + i\,\mathrm{Arg}(1+i)$$

$$= \frac{1}{2}\ln 2 + i\frac{\pi}{4}$$

$$(1+i)^{\frac{1}{2}} = \exp\left[\frac{1}{2}\left(\frac{1}{2}\ln 2 + i\frac{\pi}{4}\right)\right]$$

$$= \exp\left(\ln 2^{\frac{1}{4}} + i\frac{\pi}{8}\right)$$

$$= 2^{\frac{1}{4}}\left(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}\right.$$

$$= 1.09684 + i\,0.45509$$

Practice Problems

Problem 1: Evaluate

$\left(-\sqrt{3}-i\right)^{3}$, and $\left(-\sqrt{3}-i\right)^{-6}$

Problem 2: Prove $u(x,y)$ does not have a limit as (x,y) approaches $(0,0)$.

$$u(x,y) = \frac{xy}{x^2+y^2}$$

Problem 3: Prove $f(z)$ is differentiable only at points that lie on coordinate axis

$$f(z) = x^3 + 3xy^2 + i(y^3 + 3x^2y)$$

Problem 4: Prove, if $f(z)$ is differentiable at z_0, then is continuous at z_0